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B.Tech. Degree VII Semester Supplementary Examination April 2018

ME 1703 MACHINE DESIGN II

(2012 Scheme)

Time : 3 Hours Maximum Marks : 100

(Use of approved Design Data Handbook is permitted)

PART A
(Answer *ALL* questions)

($8 \times 5 = 40$)

I. (a) State the different classification of brakes and give at least one practical application for each.

(b) Give five advantages of V-belts over flat belts with supporting reasons.

(c) State law of gearing and derive the expression.

(d) What is interference in gears? Explain.

(e) What are the desirable properties of bearing materials?

(f) Explain the theory of hydrodynamic lubrication.

(g) What are the basic principles of designing for economic production?

(h) How are working drawings prepared for manufacturing?

PART B

($4 \times 15 = 60$)

II. A cone clutch is used to connect an electric motor transmitting 5.5 kW running at 1440 rpm. The semi cone angle is 12.5°. The mean radius of the clutch is twice as the face width. The coefficient of friction is 0.2 and the normal intensity of pressure between the contacting surfaces not to exceed 0.1 N/mm^2. Assuming the uniform wear criterion, calculate:

(i) The inner and outer diameters;
(ii) The face width of the friction lining;
(iii) The force required to engage the clutch.

OR

III. Determine the main dimensions and the capacity of double block brake for the following conditions:

Weight of hoist with load	=	2.5 kN
Downward velocity	=	1.25 m/s
Pitch diameter of hoist drum	=	1.20 m
Stopping distance for hoist	=	3 m

Neglect the kinetic energy of the drum.

IV. A reciprocating compressor is to be connected to an electric motor with the help of spur gears. The distance between the shafts is to be 500 mm. The speed of electric motor is 900 rpm and speed of the compressor shaft is described to be 200 rpm. The torque to be transmitted is 5000 Nm. Taking starting torque as 25% more than the normal torque, determine :

(i) Module and face width of the gear using 20° stub teeth.

(ii) Number of teeth and pitch circle diameter of each gear. Assume suitable value of velocity factor and Lewis equation.

OR

V. A pair of helical gears are to transmit 15 kW. The teeth are 20° stub in diametral plane and have a helix angle of 45°. The pinion runs at 1000 rpm and has 80 mm pitch diameter. The gear has 320 mm pitch diameter. If the gears are made of cast steel having allowable static strength of 100 MPa; determine a suitable module and face width from static strength considerations and check the gear for wear. Given $\sigma_{es} = 618\,MPa.$

VI. Design a journal bearing for a centrifugal pump running at 1440 rpm. Diameter of the journal is 10 cm and load on each bearing is 20 kN. Take ZN/P as 28, where N is in rpm, Z is absolute viscosity in Ns/m^2, P is bearing pressure in MN/m^2.

OR

VII. A 40 B C03 bearing is to operate in the following work cycle.

(i) Radial load (steady) of 6 kN at 200 rpm for 25% of the time.

(ii) Radial load (moderate shock) of 9 kN at 500 rpm for 20% of the time.

(iii) Radial load (light shock) of 4 kN at 400 rpm for 55% of the time.

Find the expected average life of the bearing in hours if the inner race is rotating.

VIII. What are the general design recommendations for (i) castings and (ii) forgings.

OR

IX. Explain the term "Design for assembling and disassembling (DFA/DFD) and evaluate its relevance in the context of Green Manufacturing"

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